

A Common-Hardy Head/Counterloop Rank Lock and a Hidden-Shell Obstruction

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August 2026

Abstract

We lock the actual modulus-complete noisy head and the finite graded counterloop in the same Hardy normalization and on the physical first-alias clock. The actual head consists of the algebraic nonperipheral eigenvalues of K_σ/r_H with modulus greater than $q = 1/2$; the counterloop is the $2k$ th-root shell with the two real roots removed, hence has rank $2k - 2$. Their finite genus-one quotient has local logarithmic coefficients $-d_{\sigma,k,n}/n$ for $n \geq 2$. This is an analytic comparison, not a physical determinant decomposition.

We prove two exact finite-spectral theorems. First, zero-padded root ℓ^1 matching has a cardinality floor: its cost is at least $q(r_\sigma - 2k + 2)_+ + \beta_k(2k - 2 - r_\sigma)_+$. Thus any actual matching cost tending to zero would force the eventual exact rank law $r_\sigma = 2k - 2$. No such law is presently sourced. Second, two nonzero finite multisets of ranks at most N are identical if their power sums agree for orders 2 through $2N + 1$. The missing first moment causes no ambiguity once the rank cap is known.

The cap is indispensable. Adding a complete $4k$ th-root shell of radius $3/4$ preserves every strict-prefix moment $2 \leq n < 4k$, while the padded root cost is at least $2k$ and the added genus-one factor is $1 - (3z/4)^{4k}$. This is a conjugation-closed finite normal information-class counterexample, not an actual noisy operator. Finally, the RH-299 transport criterion at the cut $4k$ requires the source-safe global rate $\gamma > 1.926813889034\dots$; the smaller $0.926813889034\dots$ threshold assumes an unproved local actual-head cap. Root transport without a rank law, cap, and actual matching rate is therefore **STOP_SCOPED**. Aggregate and annular routes remain **NOT_TESTABLE/open**, and no Gate or Riemann-hypothesis claim follows.

1 One physical data type and one clock

Set

$$r_H = \frac{17}{20}, \quad q = \frac{1}{2}, \quad R = \frac{7}{5}, \quad A_\sigma = K_\sigma/r_H. \quad (1)$$

After removing the Perron and negative-parity eigenvalues, let

$$\mathcal{H}_\sigma = \{\mu : \mu \text{ is an algebraic eigenvalue of } A_\sigma, |\mu| > q\}, \quad r_\sigma = \#\mathcal{H}_\sigma, \quad (2)$$

where algebraic multiplicity is retained, and put

$$h_{\sigma,n} = \sum_{\mu \in \mathcal{H}_\sigma} \mu^n. \quad (3)$$

This is exactly the modulus-complete head of RH-282 [7]. Its available rank information is only

$$r_\sigma \leq 4\sigma^{-1}. \quad (4)$$

In particular, the half-logarithmic endpoint singular-value rank of RH-16 [9] is not identified with r_σ . The former is a singular-resolution count; the latter is an algebraic spectral count.

Let $k = k_\sigma \geq 2$ be the physical first-alias clock reviewed in RH-341 [1]:

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad u = 4k. \quad (5)$$

For the boundary multiplier radius β_k , define the counterloop multiset

$$\mathcal{Y}_k = \{\beta_k e^{ij\pi/k}, \beta_k e^{-ij\pi/k} : 1 \leq j \leq k-1\}. \quad (6)$$

It has rank

$$m_k = \#\mathcal{Y}_k = 2k - 2, \quad (7)$$

and $\beta_k \rightarrow \beta = (r_H \sqrt{\lambda})^{-1} > q$ [4]. Since K_σ is a strongly positive Markov operator, its nonperipheral eigenvalues have modulus below one. Therefore every member of $\mathcal{H}_\sigma$ has modulus at most $1/r_H$. Also $\beta < 1/r_H$, so $\beta_k < 1/r_H$ eventually. Thus

$$B = 1/r_H \quad (8)$$

is a source-safe eventual common cap for the actual head and counterloop.

Proposition 1.1 (Exact counterloop ledger). *For every integer $n \geq 1$,*

$$s_{k,n} := \sum_{\nu \in \mathcal{Y}_k} \nu^n = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n). \quad (9)$$

The strict prefix $2 \leq n < 4k$ contains the first alias $n = 2k$ and excludes the second alias $n = 4k$.

Proof. The complete set of scaled $2k$ th roots has power sum $2k\beta_k^n \mathbf{1}_{2k|n}$. Removing the roots β_k and $-\beta_k$ subtracts $\beta_k^n(1 + (-1)^n)$, which proves (9). The two alias assertions are the strict inequalities $2 \leq 2k < 4k$ and $4k \not< 4k$. $\square$

This is the counterloop used in the typed RH-290 and RH-297 ledgers [2, 8]. Define the common-Hardy head defect

$$d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}. \quad (10)$$

The point of the source lock is that neither the RH-16 singular rank nor a model cloud may be substituted for the actual multiset (2).

2 The exact analytic factor interface

For a finite multiset X of nonzero complex numbers, write

$$E_1(w) = (1 - w)e^w, \quad \Phi_X(z) = \prod_{x \in X} E_1(xz). \quad (11)$$

The product is entire. Its logarithm has a canonical germ at the origin, since $\Phi_X(0) = 1$.

Theorem 2.1 (Common-Hardy genus-one comparison). *For the actual head $\mathcal{H}_\sigma$ and counterloop $\mathcal{Y}_k$,*

$$\log \frac{\Phi_{\mathcal{H}_\sigma}(z)}{\Phi_{\mathcal{Y}_k}(z)} = - \sum_{n \geq 2} \frac{d_{\sigma,k,n}}{n} z^n \quad (12)$$

as a convergent power series in a sufficiently small neighborhood of $z = 0$.

Proof. For $|w| < 1$,

$$\log E_1(w) = \log(1 - w) + w = - \sum_{n \geq 2} \frac{w^n}{n}. \quad (13)$$

Both multisets are finite, so one may choose a common neighborhood of zero on which (13) holds for every root. Sum first over $\mathcal{H}_\sigma$, then subtract the sum over $\mathcal{Y}_k$, and use (10). $\square$

Remark 2.2 (Claim boundary). Theorem 2.1 compares two finite analytic factors. It does not assert that the quotient is a factor in one physical determinant, that $\mathcal{Y}_k$ is a spectral submultiset of A_σ , or that the remaining physical determinant pieces glue. Those distinctions are the missing glue in RH-288 and RH-290 [5, 8].

3 Zero padding forces a rank law

If $X = \{x_1, \dots, x_r\}$ and $Y = \{y_1, \dots, y_m\}$ are finite multisets, pad the smaller one with zeros to the common size $N = \max(r, m)$ and define

$$d_1^0(X, Y) = \min_{\pi \in S_N} \sum_{j=1}^N |\tilde{x}_j - \tilde{y}_{\pi(j)}|. \quad (14)$$

This is the root interface introduced in RH-299 [6].

Theorem 3.1 (Zero-padding rank lock). *Let X be a finite multiset with $|x| > q$ for every $x \in X$, and let Y be a finite multiset whose every member has modulus $\beta_0 > 0$. If $r = \#X$ and $m = \#Y$, then*

$$d_1^0(X, Y) \geq q(r - m)_+ + \beta_0(m - r)_+. \quad (15)$$

In particular,

$$d_1^0(\mathcal{H}_\sigma, \mathcal{Y}_k) \geq q(r_\sigma - m_k)_+ + \beta_k(m_k - r_\sigma)_+. \quad (16)$$

Proof. Suppose first that $r > m$. The padded copy of Y contains exactly $r - m$ zeros. Every matching pairs those zeros with $r - m$ members of X , and each such pair costs $|x| > q$. Hence the total cost is at least $q(r - m)$. If $m > r$, the padded copy of X contains $m - r$ zeros. Exactly $m - r$ members of Y meet those zeros, each at cost β_0 , giving $\beta_0(m - r)$. The equal-rank case has the asserted zero lower bound. Apply the result with $X = \mathcal{H}_\sigma$, $Y = \mathcal{Y}_k$, and $\beta_0 = \beta_k$. $\square$

Corollary 3.2 (Any decaying actual matching locks the rank). *If*

$$d_1^0(\mathcal{H}_\sigma, \mathcal{Y}_{k_\sigma}) = o(1), \quad (17)$$

then, for all sufficiently small σ ,

$$r_\sigma = 2k_\sigma - 2. \quad (18)$$

The conclusion also follows from any estimate $d_1^0 = O(\sigma^\gamma)$ with $\gamma > 0$.

Proof. Since $\beta_k \rightarrow \beta > q$, both possible nonzero rank-mismatch floors in (16) are bounded below by a fixed positive constant for large k . An integer rank mismatch is therefore incompatible with (17). A positive power of σ tends to zero. $\square$

The corollary is a necessary consequence of a future matching theorem; it is not such a theorem. Bound (4) is far too broad to give (18), and the repository explicitly supplies no identification between the RH-16 endpoint singular rank and r_σ .

4 Shifted moments identify roots under a rank cap

For a finite multiset X , let $p_n(X) = \sum_{x \in X} x^n$.

Theorem 4.1 (Shifted-moment uniqueness). *Let X and Y be finite nonempty multisets of nonzero complex numbers with*

$$\#X \leq N, \quad \#Y \leq N. \quad (19)$$

If

$$p_n(X) = p_n(Y) \quad (2 \leq n \leq 2N + 1), \quad (20)$$

then $X = Y$ as multisets.

Proof. Define the rational function

$$F_X(z) = \sum_{x \in X} \frac{x^2}{1 - xz} = \sum_{j \geq 0} p_{j+2}(X) z^j \quad (21)$$

near the origin, and define F_Y analogously. If $r = \#X$ and $s = \#Y$, then

$$F_X = \frac{A_X}{Q_X}, \quad Q_X(z) = \prod_{x \in X} (1 - xz), \quad \deg A_X \leq r - 1, \quad (22)$$

with the analogous bounds for Y . Consequently the numerator of $F_X - F_Y$ over $Q_X Q_Y$ has degree at most

$$\max\{(r - 1) + s, (s - 1) + r\} = r + s - 1 \leq 2N - 1. \quad (23)$$

Condition (20) says that the first $2N$ Taylor coefficients in (21) agree. Thus $F_X - F_Y$ has a zero of order at least $2N$ at the origin. Since $Q_X(0)Q_Y(0) = 1$, its polynomial numerator is divisible by z^{2N} . The degree bound (23) forces that numerator to vanish identically, so $F_X = F_Y$.

For every distinct nonzero value a occurring in X , the partial fraction of F_X contains

$$\frac{m_X(a)a^2}{1 - az}. \quad (24)$$

Equality of the rational functions gives the same poles and the same coefficients, hence $m_X(a) = m_Y(a)$ for every a . Therefore the multisets are identical. $\square$

Corollary 4.2 (Strict-prefix identification with a cap). *Let $N = 2k - 2$. If X is a finite multiset of nonzero complex numbers, has rank at most N , and*

$$p_n(X) = s_{k,n} \quad (2 \leq n < 4k), \quad (25)$$

then $X = \mathcal{Y}_k$.

Proof. The model has rank N . The required terminal order in Theorem 4.1 is $2N + 1 = 4k - 3 < 4k$, so (25) contains every required moment. $\square$

This corollary shows precisely what the absence of the first moment does *not* obstruct. Its active hypothesis is the rank cap. No such actual cap at $2k - 2$ is presently proved.

5 A hidden-shell counterexample without the cap

Let

$$\mathcal{Z}_k = \left\{ \frac{3}{4} e^{2\pi i j / (4k)} : 0 \leq j < 4k \right\}, \quad \mathcal{X}_k = \mathcal{Y}_k \sqcup \mathcal{Z}_k. \quad (26)$$

Both multisets are closed under complex conjugation. They are spectra of finite normal diagonal matrices.

Theorem 5.1 (Hidden-shell obstruction). *For every $k \geq 2$ and every order $2 \leq n < 4k$,*

$$p_n(\mathcal{X}_k) = p_n(\mathcal{Y}_k). \quad (27)$$

Thus the strict-prefix head budget formed from these two multisets is exactly zero:

$$\sum_{2 \leq n < 4k} \frac{|p_n(\mathcal{X}_k) - p_n(\mathcal{Y}_k)| R^n}{n} = 0. \quad (28)$$

For all sufficiently large k ,

$$d_1^0(\mathcal{X}_k, \mathcal{Y}_k) \geq 4kq = 2k. \quad (29)$$

Moreover,

$$\frac{\Phi_{\mathcal{X}_k}(z)}{\Phi_{\mathcal{Y}_k}(z)} = 1 - \left(\frac{3z}{4} \right)^{4k}. \quad (30)$$

Proof. The complete root shell satisfies

$$\sum_{\zeta \in \mathcal{Z}_k} \zeta^n = 4k \left(\frac{3}{4}\right)^n \mathbf{1}_{4k|n}. \quad (31)$$

It vanishes throughout $2 \leq n < 4k$, which proves (27) and (28).

The rank difference is $\#\mathcal{X}_k - \#\mathcal{Y}_k = 4k$. For large k , every member of $\mathcal{Y}_k$ has modulus $\beta_k > q$, and every member of $\mathcal{Z}_k$ has modulus $3/4 > q$. Apply the first branch of Theorem 3.1 to obtain (29).

Finally, the genus-one exponential terms cancel because $\sum_{\zeta \in \mathcal{Z}_k} \zeta = 0$. The elementary full-root product gives

$$\prod_{\zeta \in \mathcal{Z}_k} (1 - \zeta z) = 1 - (3z/4)^{4k}, \quad (32)$$

which is (30). $\square$

Remark 5.2 (Exact scope of the counterexample). Theorem 5.1 is a finite normal spectral information-class example. It proves that strict-prefix moments do not control padded root distance without a rank cap, and that an entire shell may first appear at the excluded endpoint. It does not realize K_σ , prove a physical head-rank mismatch, show that the actual D_{4k} diverges, or refute a future signed aggregate theorem.

6 The RH-299 threshold on the physical cut

RH-299 proves that if both root multisets lie in $|z| \leq B$, then for every integer $m \geq 3$ [6]

$$D_m(R) := \sum_{n=2}^{m-1} \frac{|p_n(X) - p_n(Y)| R^n}{n} \leq d_1^0(X, Y) R \sum_{j=1}^{m-2} (BR)^j. \quad (33)$$

Proposition 6.1 (Physical-clock rate thresholds). *Put $m = 4k$ with the clock (5), and suppose $BR > 1$ and*

$$d_1^0(\mathcal{H}_\sigma, \mathcal{Y}_k) = O(\sigma^\gamma). \quad (34)$$

Then the right side of (33) tends to zero if

$$\gamma > \frac{2 \log(BR)}{\log \lambda}. \quad (35)$$

For the source-safe common cap $B = 1/r_H$, this becomes

$$\gamma > \frac{2 \log(R/r_H)}{\log \lambda} = 1.926813889034004 \dots \quad (36)$$

If one had the additional local actual-head cap

$$|\mu| \leq \beta + o(1) \quad (\mu \in \mathcal{H}_\sigma), \quad (37)$$

then the limiting threshold would instead be

$$\gamma > \frac{2 \log(\beta R)}{\log \lambda} = 0.926813889034004 \dots \quad (38)$$

The local cap is not supplied by the repository.

Proof. Since $BR > 1$, the geometric sum in (33) is $O((BR)^m)$. Equations (5) and $m = 4k$ give

$$m = \frac{2 \log(1/\sigma)}{\log \lambda} + O(1), \quad (BR)^m = O\left(\sigma^{-2 \log(BR)/\log \lambda}\right). \quad (39)$$

Multiplication by (34) proves (35). Substituting $B = 1/r_H$ gives (36). Substituting $\beta = (r_H \sqrt{\lambda})^{-1}$ gives the limiting value in (38). More precisely, for every γ strictly above that value, choose $\varepsilon > 0$ so small that $\gamma > 2 \log((\beta + \varepsilon)R)/\log \lambda$. Equation (37) and $\beta_k \rightarrow \beta$ then give the common eventual cap $B = \beta + \varepsilon$. The two displayed limiting thresholds differ by exactly one, because

$$\frac{2}{\log \lambda} \log \frac{R/r_H}{\beta R} = \frac{2}{\log \lambda} \log \sqrt{\lambda} = 1. \quad (40)$$

□

The specialization is only a sufficient interface. Even (36) requires an actual matching theorem, and Corollary 3.2 shows that such a rate also contains the eventual exact rank law. The smaller number in (38) cannot be quoted for the actual head before proving the local cap.

7 Protocol, verdict, and next admissible input

The executable artifact uses exact rational constants for q , r_H , R , and the hidden radius. It evaluates the counterloop and full-shell power formulas, the rank-mismatch floor, the rational-function degree ledger, and the two RH-299 thresholds. Finite rows at $k = 3, 5, 9, 17$ are reproduction checks only; they are not noisy spectral observations and have no asymptotic evidentiary role.

The results separate three conclusions:

1. The rank-lock and shifted-moment theorems are exact finite-spectral statements in the declared common-Hardy data type.
2. The hidden shell proves a strict cap-free information-class obstruction, not a physical nonidentification theorem.
3. No source supplies an actual rank law, local shell cap, root list, or matching rate. Therefore RH-299 activation without these hypotheses is **STOP_SCOPED**.

Aggregate moment, Fourier, and Hardy routes remain **NOT_TESTABLE/open**: the counterexample prevents a cap-free logical upgrade to root matching but does not decide the actual aggregate defect. The direct annular route of RH-300 [3] also remains **NOT_TESTABLE/open** because no theorem controls

$$g_\sigma(z) = \sum_{n \geq 2} (\tau_{\sigma,n} - a_n) \frac{z^n}{n} \quad (41)$$

on any annulus $1.4 < \rho < r_H \lambda$. The narrow reopen inputs are an actual head-rank theorem, an actual root matching with a proved common cap and sufficient rate, or direct annular convergence for (41).

No determinant gluing is activated, and Gates A–E remain false/open. This paper proves neither physical head nonidentification nor physical defect divergence. It constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace or completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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